

S-38.3159 Protocol Design: Final evaluation of implementation

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Chapter 1

Evaluation

This report will assemble all of test results for IoT pub/sub protocol implementation. This report will be decomposed into six parts; Test one to six. Task one and two will be studied in ideal scenario, Task three and four will focus on reliability issues. The report will present freshness, delay and loss metric. In tasks five and six we will study the congestion and its effect on the protocol performance. Our implementation have own Markovian Packet Loss Generator.

1.1 Test 1

Scenario: 1 client connect to all sensors

Environment: No losses, delays or b/w restrictions

Client	Camera	Device	Temperature	GPS
Mean Delay	0	0	0	0
Max Delay	0	0	0	0
Min Delay	0	0	0	0
Variance Delay	0	0	0	0

Table 1.1: Scheduling and Load Distribution.

1.2 Test 2

Scenario: N client connect to all sensors

Environment: No losses, delays or b/w restrictions

Test 2					
Metric	Client	Camera	Device	Temperature	GPS
Mean Delay	client 1	0.0008139527	0	0.000333333	0
	client 2	0.0005813948	0	0.000333333	0
	client 3	0.000116279	0	0.000333333	0
	client 4	0.000116279	0	0.000333333	0
Max Delay	client 1	0.00999999	0	0.00999999	0
	client 2	0.00999999	0	0.00999999	0
	client 3	0.00999999	0	0.00999999	0
	client 4	0.00999999	0	0.00999999	0
Min Delay	client 1	0	0	0	0
	client 2	0	0	0	0
	client 3	0	0	0	0
	client 4	0	0	0	0
Variance Delay	client 1	7.564965e-06	0	3.333327e-06	0
	client 2	5.540345e-06	0	3.333327e-06	0
	client 3	1.162788e-06	0	3.333327e-06	0
	client 4	1.162788e-06	0	3.333327e-06	0

1.3 Test 3: Reliability

Scenario: 4 clients connect to all sensors

Environment: Each client has a different loss rate 0%, 1%, 10%,33% and markov chain model for packet loss.

Configuration:

#delay(ms)	p_prob	q_prob
0	0.0	0.5
0	0.01	0.5
0	0.10	0.5
0	0.33	0.5

Test 3 with Unreliable					
Metric	Client	Camera	Device	Temperature	GPS
Mean Delay	client 1	0.0004687496	0	0	0
	client 2	0.0003124997	0	0	0
	client 3	0.0002343748	1.001379	1.001379	0
	client 4	0	2.269231	1.001379	0
Max Delay	client 1	0.00999999	0	0	0
	client 2	0.00999999	0	0	0
	client 3	0.00999999	4.06	1.01	0
	client 4	0	4.06	1.001379	0
Min Delay	client 1	0	0	0	0
	client 2	0	0	0	0
	client 3	0	1.06	1	0
	client 4	0	1.06	1	0
Variance Delay	client 1	3.2258e-06	0	3.33327e-06	0
	client 2	3.2258e-06	0	0	0
	client 3	3.2258e-06	1.014627	1.231525e-05	0
	client 4	3.2258e-06	1.014627	1.231525e-05	0
Mean Skew	client 1	0	0	0	0
	client 2	0	0	0	0
	client 3	0	1	1	0
	client 4	0	1	1	0
Max Skew	client 1	0	0	0	0
	client 2	0	0	0	0
	client 3	0	1	1	0
	client 4	0	1	1	0
Min Skew	client 1	0	0	0	0
	client 2	0	0	0	0
	client 3	0	1	1	0
	client 4	0	1	1	0
Packet Loss	client 1	0	0	0	0
	client 2	0	1	0	0
	client 3	0	1	1	0
	client 4	0	1	1	0

Test 3 with reliable					
Metric	Client	Camera	Device	Temperature	GPS
Mean Delay	client 1	0.0003409088	0.0006666666	0.0006666666	0
	client 2	0.0003409088	0.0006666666	0.0006666666	0
	client 3	0.0001136363	0.0006666666	1.002069	7.06
	client 4	0.0002272725	0.0006666666	1.002069	7.06
Max Delay	client 1	0.00999999	0.00999999	0	0
	client 2	0.00999999	0.00999999	0	0
	client 3	0.00999999	0.00999999	1.01	9.19
	client 4	0.00999999	0.00999999	1.01	9.19
Min Delay	client 1	0	0	0	0
	client 2	0	0	0	0
	client 3	0	0	4.36	0
	client 4	0	0	4.36	0
Variance Delay	client 1	3.330715e-06	6.666654e-06	6.436769e-06	0
	client 2	3.330715e-06	6.666654e-06	6.436769e-06	0
	client 3	1.136361e-06	6.666654e-06	1.699504e-05	6.0759
	client 4	2.2466e-06	6.666654e-06	1.699504e-05	6.0759
Mean Skew	client 1	0	0	0	0
	client 2	0	0	0	0
	client 3	0	0	0	1
	client 4	0	0	0	1
Max Skew	client 1	0	0	0	0
	client 2	0	0	0	0
	client 3	0	0	0	1
	client 4	0	0	0	1
Min Skew	client 1	0	0	0	0
	client 2	0	0	0	0
	client 3	0	0	0	1
	client 4	0	0	0	1
Skew Variance	client 1	0	0	0	0
	client 2	0	0	0	0
	client 3	0	0	0	0
	client 4	0	0	0	0
Packet Loss	client 1	0	0	0	0
	client 2	0	0	0	0
	client 3	0	0	1	1
	client 4	0	0	1	1

1.4 Test 4: Reliability

Scenario: 4 clients connect to all sensors

Environment: Each client has a different path latency 0%, 25%, 100%, 400% and markov chain model for packet loss.

Configuration:

#delay(ms)	p_prob	q_prob
0	0.0	1.0
25	0.0	1.0
100	0.0	1.0
400	0.0	1.0

Test 4 with Unreliable					
Metric	Client	Camera	Device	Temperature	GPS
Mean Delay	client 1	0.0003225803	0	0.0006666666	0
	client 2	0.0003225803	2.269231	0.0003333333	0
	client 3	0.0003225803	2.269231	1.001379	0
	client 4	0.0003225803	2.269231	1.001379	0
Max Delay	client 1	0.00999999	0	0.00999999	0
	client 2	0.00999999	4.56	0.00999999	0
	client 3	0.00999999	4.56	1.01	0
	client 4	0.00999999	4.56	1.01	0
Min Delay	client 1	0	0	0	0
	client 2	0	0.5699999	0	0
	client 3	0	0.5699999	0	0
	client 4	0	0.5699999	0	0
Variance Delay	client 1	3.2258e-06	0	0	0
	client 2	3.2258e-06	1.480958	0	0
	client 3	3.2258e-06	1.480958	1.231525e-05	0
	client 4	3.2258e-06	1.480958	1.231525e-05	0
Mean Skew	client 1	0	0	0	0
	client 2	0	1	0	0
	client 3	0	1	1	0
	client 4	0	1	1	0
Max Skew	client 1	0	0	0	0
	client 2	0	1	0	0
	client 3	0	1	1	0
	client 4	0	1	1	0
Min Skew	client 1	0	0	0	0
	client 2	0	0	0	0
	client 3	0	0	0	0
	client 4	0	0	0	0
Packet Loss	client 1	0	0	0	0
	client 2	0	1	0	0
	client 3	0	1	1	0
	client 4	0	1	1	0

Test 4 with reliable					
Metric	Client	Camera	Device	Temperature	GPS
Mean Delay	client 1	0.0009433953	0.00142857	0.00033333	0
	client 2	0.0005660372	0.00142857	0.00033333	0
	client 3	0.0005660372	2.292308	0.00033333	0
	client 4	0.0005660372	2.292308	0.00033333	0
Max Delay	client 1	0.00999999	0.00999999	0.00999999	0
	client 2	0.00999999	0.00999999	0.00999999	0
	client 3	0.00999999	4.54	0.00999999	0
	client 4	0.00999999	4.54	0.00999999	0
Min Delay	client 1	0	0	0	0
	client 2	0	0	0	0
	client 3	0	0	0	0
	client 4	0	0	0	0
Variance Delay	client 1	8.708256e-06	1.318679e-05	3.333327e-06	0
	client 2	5.44266e-06	1.318679e-05	3.333327e-06	0
	client 3	5.44266e-06	1.792886	3.333327e-06	0
	client 4	5.44266e-06	1.792886	3.333327e-06	0
Mean Skew	client 1	0	0	0	0
	client 2	0	0	0	0
	client 3	0	1	0	0
	client 4	0	1	0	0
Max Skew	client 1	0	0	0	0
	client 2	0	0	0	0
	client 3	0	1	0	0
	client 4	0	1	0	0
Min Skew	client 1	0	0	0	0
	client 2	0	0	0	0
	client 3	0	1	0	0
	client 4	0	1	0	0
Skew Variance	client 1	0	0	0	0
	client 2	0	0	0	0
	client 3	0	0	0	0
	client 4	0	0	0	0
Packet Loss	client 1	0	0	0	0
	client 2	0	0	0	0
	client 3	0	1	0	0
	client 4	0	1	0	0

1.5 Test 5: Congestion control

In tasks five and six we study the congestion and its effect on the protocol performance. In order to provide reliable transport while not contributing to congestion collapse many protocols provide or use different mechanisms for congestion control and flow control. In this section we study the effect of heterogeneous bandwidth in different configurations our protocol implementation supports.

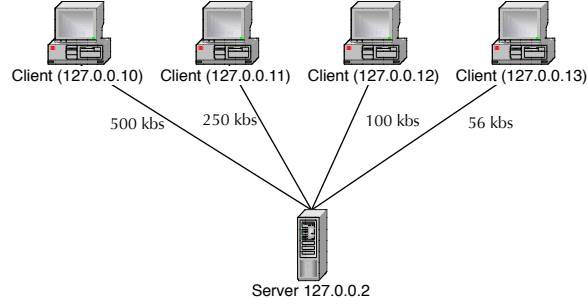


Figure 1.1: Number of flows with given flow timeout values

client	mean delay	max delay	min delay	variance delay	lost messages (out of 88)
1 (500kbs)	0.0228	0.0399	0	0.0001	0
2 (250kbs)	0.0443	0.0800	0	0.0003	0
3 (100kbs)	0.7598	1.56	0.0090	0.2056	0
4 (56kbs)	1.9263	3.96	0.0199	1.4433	0.03

Table 1.2: Receive statistics for camera sensor data (no realibility guarantees, no congestion control)

1.5.1 Evaluation scenario

To evaluate the performance under the assumptions given in this task four clients were assigned a IP address from loopback address block (127.0.0.0/8). The server belongs also to this address block. This network setup of ours is given in figure 1.1. These additional addresses were created as alises with command:

```
sudo ifconfig lo0 alias 127.0.0.2
```

Dummynet was used to create link characteristics between client and server machines with e.g. following command for each link between client and server (naturally changing the IP addresses as required):

```
sudo ipfw add pipe 1 ip from 127.0.0.2 to 127.0.0.10;
sudo ipfw pipe 1 config bw 500kb
```

1.5.2 No weak-reliability, no congestion control

This evaluation gives basically a base delays for different

client	mean delay	max delay	min delay	variance delay	lost messages (out of 88)
1 (500kbs)	0.0226	0.09	0	0.0001	0
2 (250kbs)	0.0428	0.0900	0	0.0003	0
3 (100kbs)	0.5742	1.56	0.0199	0.1305	0
4 (56kbs)	1.6785	4.05	0.0299	1.1156	0.02

Table 1.3: Receive statistics for camera sensor data (no realibility guarantees, skipping messages)

1.5.3 No weak-reliability, skipping messages

1.5.4 Weak-reliability, no congestion control

1.5.5 Weak-realiability, skipping messages

1.6 Test 6: Congestion control and Reliability

In this section we study the effect of heterogeneous bandwidth with different packet loss rates in different configurations our protocol implementation supports.

1.6.1 Evaluation scenario

To evaluate the performance under the assumptions given in this task four clients were assigned a IP address from loopback address block (127.0.0.0/8). The server belongs also to this address block. This network setup of ours is given in figure 1.1. These additional addresses were created as alises with command:

```
sudo ifconfig lo0 alias 127.0.0.2
```

Dummynet was used to create link characteristics between client and server machines with e.g. following command for each link between client and server (naturally changing the IP addresses as required):

```
sudo ipfw add pipe 1 ip from 127.0.0.2 to 127.0.0.10;  
sudo ipfw pipe 1 config bw 500kb
```

The packet loss was created with our transport layer implementation which is capable to simulate artificial packet loss and delays. It's good to notice however that this is not the most realistic situation as packet loss is only happening on the link from server to client. This means that when a weakly realiable subscriptions are used all acknowledgements are not experiencing any packet loss except in case that happens naturally.

client (bw, plr)	mean delay	max delay	min delay	variance delay	packet loss
1 (500kbs, 0.05)	0.0418	0.0799	0	0.0003	0.22
2 (500kbs, 0.15)	0.0403	0.1000	0	0.0006	0.71
3 (100kbs, 0.05)	1.6777	2.9300	0.01999	0.7091	0.37
4 (100kbs, 0.15)	1.3811	2.4500	0.0299	0.5699	0.56

Table 1.4: Task 6: Receive statistics for camera sensor data (no realibility guarantees, no congestion control)

client	mean delay	max delay	min delay	variance delay	packet loss
1 (500kbs, 0.05)	0.0412	0.0799	0	0.0004	0.34
2 (500kbs, 0.15)	0.0438	0.0799	0	0.0004	0.45
3 (100kbs, 0.05)	1.7038	2.6500	0.0299	0.9063	0.34
4 (100kbs, 0.15)	1.2141	2.65	0.0299	0.6901	0.44

Table 1.5: Task 6: Receive statistics for camera sensor data (no realibility guarantees, skipping messages)

1.6.2 Results

The scenario where we have no weak-reliability guarantees or active congestion control are shown in table 1.4 for camera sensor data. We observe quite high packet loss rate due to fragmentation while also the effect of bandwidth is clearly shown.

The statistics when message skipping is turned on are shown in table 1.5. The results show slightly decreased delays for low bandwidth clients while the effect for delays of high bandwidth clients is negligible.

1.6.3 Weak-reliability, no congestion control

1.6.4 Weak-reliability, skipping messages

1.7 Lessons Learnt

We think that our implementation run quite smoothly, except for the congestion control part. We can argue that CC is quite useless in this kind of applications. Also combining implementation and specification at the same time and starting early helped us quite a lot. According to our point of view that we have implemented about 80-90% of the specification. Additionally, our CC is not so specified as it could be.

Appendices

Appendix A

Results

Test 1

```
Sensor camera_3
Server got 23 out of 23
client1 lost 0 out of 23
client1 duplicated (possibly fragments) 0
client1 out of order 0
  mean delay 0
  max delay 0
  min delay 0
  variance delay 0
```

```
-----
Sensor device_0
Server got 12 out of 12
client1 lost 0 out of 12
client1 duplicated (possibly fragments) 0
client1 out of order 0
  mean delay 0
  max delay 0
  min delay 0
  variance delay 0
```

```
-----
Sensor gps_2
Server got 5 out of 5
client1 lost 0 out of 5
client1 duplicated (possibly fragments) 0
client1 out of order 0
  mean delay 0
  max delay 0
  min delay 0
```

```

variance delay 0
-----
Sensor temp_1
Server got 30 out of 30
client1 lost 0 out of 30
client1 duplicated (possibly fragments) 0
client1 out of order 0
  mean delay 0
  max delay 0
  min delay 0
  variance delay 0
-----

```

Test 2

Scenario: N client connect to all sensors

Environment: No losses, delays or b/w restrictions

```

Sensor camera_3
Server got 86 out of 86
client1 lost 0 out of 86
client1 duplicated (possibly fragments) 0
client1 out of order 0
  mean delay 0.0008139527
  max delay 0.00999999
  min delay 0
  variance delay 7.564965e-06
client2 lost 0 out of 86
client2 duplicated (possibly fragments) 0
client2 out of order 0
  mean delay 0.0005813948
  max delay 0.00999999
  min delay 0
  variance delay 5.540345e-06
client3 lost 0 out of 86
client3 duplicated (possibly fragments) 0
client3 out of order 0
  mean delay 0.000116279
  max delay 0.00999999
  min delay 0
  variance delay 1.162788e-06
client4 lost 0 out of 86
client4 duplicated (possibly fragments) 0
client4 out of order 0

```

```

mean delay 0.000116279
max delay 0.00999999
min delay 0
variance delay 1.162788e-06
-----
Sensor device_0
Server got 13 out of 13
client1 lost 0 out of 13
client1 duplicated (possibly fragments) 0
client1 out of order 0
  mean delay 0
  max delay 0
  min delay 0
  variance delay 0
client2 lost 0 out of 13
client2 duplicated (possibly fragments) 0
client2 out of order 0
  mean delay 0
  max delay 0
  min delay 0
  variance delay 0
client3 lost 0 out of 13
client3 duplicated (possibly fragments) 0
client3 out of order 0
  mean delay 0
  max delay 0
  min delay 0
  variance delay 0
client4 lost 0 out of 13
client4 duplicated (possibly fragments) 0
client4 out of order 0
  mean delay 0
  max delay 0
  min delay 0
  variance delay 0
-----
Sensor gps_2
Server got 4 out of 4
client1 lost 0 out of 4
client1 duplicated (possibly fragments) 0
client1 out of order 0
  mean delay 0
  max delay 0
  min delay 0

```

```

    variance delay 0
client2 lost 0 out of 4
client2 duplicated (possibly fragments) 0
client2 out of order 0
    mean delay 0
    max delay 0
    min delay 0
    variance delay 0
client3 lost 0 out of 4
client3 duplicated (possibly fragments) 0
client3 out of order 0
    mean delay 0
    max delay 0
    min delay 0
    variance delay 0
client4 lost 0 out of 4
client4 duplicated (possibly fragments) 0
client4 out of order 0
    mean delay 0
    max delay 0
    min delay 0
    variance delay 0

```

```

-----
Sensor temp_1
Server got 30 out of 30
client1 lost 0 out of 30
client1 duplicated (possibly fragments) 0
client1 out of order 0
    mean delay 0.000333333
    max delay 0.00999999
    min delay 0
    variance delay 3.333327e-06
client2 lost 0 out of 30
client2 duplicated (possibly fragments) 0
client2 out of order 0
    mean delay 0.000333333
    max delay 0.00999999
    min delay 0
    variance delay 3.333327e-06
client3 lost 0 out of 30
client3 duplicated (possibly fragments) 0
client3 out of order 0
    mean delay 0.000333333
    max delay 0.00999999

```

```

min delay 0
variance delay 3.333327e-06
client4 lost 0 out of 30
client4 duplicated (possibly fragments) 0
client4 out of order 0
mean delay 0.000333333
max delay 0.00999999
min delay 0
variance delay 3.333327e-06
-----

> proc.time()
      user  system elapsed
    0.160   0.016   0.169

```

Test 4 with Unreliable

```

Sensor camera_3
Server got 31 out of 31
client1 lost 0 out of 31
client1 duplicated 0
client1 out of order 0
mean delay 0.0003225803
max delay 0.00999999
min delay 0
variance delay 3.2258e-06
mean skew 0
max skew 0
min skew 0
variance skew 0
client2 lost 0 out of 31
client2 duplicated 0
client2 out of order 0
mean delay 0.0003225803
max delay 0.00999999
min delay 0
variance delay 3.2258e-06
mean skew 0
max skew 0
min skew 0
variance skew 0
client3 lost 0 out of 31
client3 duplicated 0

```



```

client3 out of order 0
  mean delay 0.0003225803
  max delay 0.00999999
  min delay 0
  variance delay 3.2258e-06
  mean skew 0
  max skew 0
  min skew 0
  variance skew 0
client4 lost 0 out of 31
client4 duplicated 0
client4 out of order 0
  mean delay 0.0003225803
  max delay 0.00999999
  min delay 0
  variance delay 3.2258e-06
  mean skew 0
  max skew 0
  min skew 0
  variance skew 0

```

```

-----
Sensor device_0
Server got 14 out of 14
client1 lost 0 out of 14
client1 duplicated 0
client1 out of order 0
  mean delay 0
  max delay 0
  min delay 0
  variance delay 0
  mean skew 0
  max skew 0
  min skew 0
  variance skew 0
client2 lost 1 out of 14
client2 duplicated 0
client2 out of order 0
  mean delay 2.269231
  max delay 4.56
  min delay 0.5699999
  variance delay 1.480958
  mean skew 1
  max skew 1
  min skew 1

```

```
    variance skew 0
client3 lost 1 out of 14
client3 duplicated 0
client3 out of order 0
    mean delay 2.269231
    max delay 4.56
    min delay 0.5699999
    variance delay 1.480958
    mean skew 1
    max skew 1
    min skew 1
    variance skew 0
client4 lost 1 out of 14
client4 duplicated 0
client4 out of order 0
    mean delay 2.269231
    max delay 4.56
    min delay 0.5699999
    variance delay 1.480958
    mean skew 1
    max skew 1
    min skew 1
    variance skew 0
```

```
-----
Sensor gps_2
Server got 4 out of 4
client1 lost 0 out of 4
client1 duplicated 0
client1 out of order 0
    mean delay 0
    max delay 0
    min delay 0
    variance delay 0
    mean skew 0
    max skew 0
    min skew 0
    variance skew 0
client2 lost 0 out of 4
client2 duplicated 0
client2 out of order 0
    mean delay 0
    max delay 0
    min delay 0
    variance delay 0
```

```

mean skew 0
max skew 0
min skew 0
variance skew 0
client3 lost 0 out of 4
client3 duplicated 0
client3 out of order 0
mean delay 0
max delay 0
min delay 0
variance delay 0
mean skew 0
max skew 0
min skew 0
variance skew 0
client4 lost 0 out of 4
client4 duplicated 0
client4 out of order 0
mean delay 0
max delay 0
min delay 0
variance delay 0
mean skew 0
max skew 0
min skew 0
variance skew 0
-----
Sensor temp_1
Server got 30 out of 30
client1 lost 0 out of 30
client1 duplicated 0
client1 out of order 0
mean delay 0
max delay 0
min delay 0
variance delay 0
mean skew 0
max skew 0
min skew 0
variance skew 0
client2 lost 0 out of 30
client2 duplicated 0
client2 out of order 0
mean delay 0

```

```

max delay 0
min delay 0
variance delay 0
mean skew 0
max skew 0
min skew 0
variance skew 0
client3 lost 1 out of 30
client3 duplicated 0
client3 out of order 0
mean delay 1.001379
max delay 1.01
min delay 1
variance delay 1.231525e-05
mean skew 1
max skew 1
min skew 1
variance skew 0
client4 lost 1 out of 30
client4 duplicated 0
client4 out of order 0
mean delay 1.001379
max delay 1.01
min delay 1
variance delay 1.231525e-05
mean skew 1
max skew 1
min skew 1
variance skew 0
-----
>
> proc.time()
      user  system elapsed
    0.184   0.012   0.189

```

Test 4 with Reliable Parameter

```

Sensor camera_3
Server got 53 out of 53
client1 lost 0 out of 53
client1 duplicated 0
client1 out of order 0
mean delay 0.0009433953

```

```

max delay 0.00999999
min delay 0
variance delay 8.708256e-06
mean skew 0
max skew 0
min skew 0
variance skew 0
client2 lost 0 out of 53
client2 duplicated 0
client2 out of order 0
mean delay 0.0005660372
max delay 0.00999999
min delay 0
variance delay 5.44266e-06
mean skew 0
max skew 0
min skew 0
variance skew 0
client3 lost 0 out of 53
client3 duplicated 0
client3 out of order 0
mean delay 0.0005660372
max delay 0.00999999
min delay 0
variance delay 5.44266e-06
mean skew 0
max skew 0
min skew 0
variance skew 0
client4 lost 0 out of 53
client4 duplicated 0
client4 out of order 0
mean delay 0.0005660372
max delay 0.00999999
min delay 0
variance delay 5.44266e-06
mean skew 0
max skew 0
min skew 0
variance skew 0
-----
Sensor device_0
Server got 14 out of 14
client1 lost 0 out of 14

```

```
client1 duplicated 0
client1 out of order 0
  mean delay 0.00142857
  max delay 0.00999999
  min delay 0
  variance delay 1.318679e-05
  mean skew 0
  max skew 0
  min skew 0
  variance skew 0
client2 lost 0 out of 14
client2 duplicated 0
client2 out of order 0
  mean delay 0.00142857
  max delay 0.00999999
  min delay 0
  variance delay 1.318679e-05
  mean skew 0
  max skew 0
  min skew 0
  variance skew 0
client3 lost 1 out of 14
client3 duplicated 0
client3 out of order 0
  mean delay 2.292308
  max delay 4.54
  min delay 0.1899998
  variance delay 1.792886
  mean skew 1
  max skew 1
  min skew 1
  variance skew 0
client4 lost 1 out of 14
client4 duplicated 0
client4 out of order 0
  mean delay 2.292308
  max delay 4.54
  min delay 0.1899998
  variance delay 1.792886
  mean skew 1
  max skew 1
  min skew 1
  variance skew 0
-----
```

Sensor gps_2
Server got 5 out of 5
client1 lost 0 out of 5
client1 duplicated 0
client1 out of order 0
mean delay 0
max delay 0
min delay 0
variance delay 0
mean skew 0
max skew 0
min skew 0
variance skew 0
client2 lost 0 out of 5
client2 duplicated 0
client2 out of order 0
mean delay 0
max delay 0
min delay 0
variance delay 0
mean skew 0
max skew 0
min skew 0
variance skew 0
client3 lost 0 out of 5
client3 duplicated 0
client3 out of order 0
mean delay 0
max delay 0
min delay 0
variance delay 0
mean skew 0
max skew 0
min skew 0
variance skew 0
client4 lost 0 out of 5
client4 duplicated 0
client4 out of order 0
mean delay 0
max delay 0
min delay 0
variance delay 0
mean skew 0
max skew 0

```

min skew 0
variance skew 0
-----
Sensor temp_1
Server got 30 out of 30
client1 lost 0 out of 30
client1 duplicated 0
client1 out of order 0
  mean delay 0.000333333
  max delay 0.00999999
  min delay 0
  variance delay 3.333327e-06
  mean skew 0
  max skew 0
  min skew 0
  variance skew 0
client2 lost 0 out of 30
client2 duplicated 0
client2 out of order 0
  mean delay 0.000333333
  max delay 0.00999999
  min delay 0
  variance delay 3.333327e-06
  mean skew 0
  max skew 0
  min skew 0
  variance skew 0
client3 lost 0 out of 30
client3 duplicated 0
client3 out of order 0
  mean delay 0.000333333
  max delay 0.00999999
  min delay 0
  variance delay 3.333327e-06
  mean skew 0
  max skew 0
  min skew 0
  variance skew 0
client4 lost 0 out of 30
client4 duplicated 0
client4 out of order 0
  mean delay 0.000333333
  max delay 0.00999999
  min delay 0

```



```

variance delay 3.333327e-06
mean skew 0
max skew 0
min skew 0
variance skew 0
-----
>
> proc.time()
      user  system elapsed
    0.216   0.016   0.250

```

Test 3 with Reliable parameter

```

Sensor camera_3
Server got 88 out of 88
client1 lost 0 out of 88
client1 duplicated 0
client1 out of order 0
  mean delay 0.0003409088
  max delay 0.009999999
  min delay 0
  variance delay 3.330715e-06
  mean skew 0
  max skew 0
  min skew 0
  variance skew 0
client2 lost 0 out of 88
client2 duplicated 0
client2 out of order 0
  mean delay 0.0003409088
  max delay 0.009999999
  min delay 0
  variance delay 3.330715e-06
  mean skew 0
  max skew 0
  min skew 0
  variance skew 0
client3 lost 0 out of 88
client3 duplicated 0
client3 out of order 0
  mean delay 0.0001136363
  max delay 0.009999999
  min delay 0

```

variance delay 1.136361e-06
mean skew 0
max skew 0
min skew 0
variance skew 0
client4 lost 0 out of 88
client4 duplicated 0
client4 out of order 0
mean delay 0.0002272725
max delay 0.00999999
min delay 0
variance delay 2.2466e-06
mean skew 0
max skew 0
min skew 0
variance skew 0

Sensor device_0
Server got 15 out of 15
client1 lost 0 out of 15
client1 duplicated 0
client1 out of order 0
mean delay 0.000666666
max delay 0.00999999
min delay 0
variance delay 6.66654e-06
mean skew 0
max skew 0
min skew 0
variance skew 0
client2 lost 0 out of 15
client2 duplicated 0
client2 out of order 0
mean delay 0.000666666
max delay 0.00999999
min delay 0
variance delay 6.66654e-06
mean skew 0
max skew 0
min skew 0
variance skew 0
client3 lost 0 out of 15
client3 duplicated 0
client3 out of order 0

```
mean delay 0.000666666
max delay 0.00999999
min delay 0
variance delay 6.66654e-06
mean skew 0
max skew 0
min skew 0
variance skew 0
client4 lost 0 out of 15
client4 duplicated 0
client4 out of order 0
mean delay 0.000666666
max delay 0.00999999
min delay 0
variance delay 6.66654e-06
mean skew 0
max skew 0
min skew 0
variance skew 0
```

```
-----
Sensor gps_2
Server got 4 out of 4
client1 lost 0 out of 4
client1 duplicated 0
client1 out of order 0
mean delay 0
max delay 0
min delay 0
variance delay 0
mean skew 0
max skew 0
min skew 0
variance skew 0
client2 lost 0 out of 4
client2 duplicated 0
client2 out of order 0
mean delay 0
max delay 0
min delay 0
variance delay 0
mean skew 0
max skew 0
min skew 0
variance skew 0
```

client3 lost 1 out of 4
client3 duplicated 0
client3 out of order 0
mean delay 7.06
max delay 9.19
min delay 4.36
variance delay 6.0759
mean skew 1
max skew 1
min skew 1
variance skew 0
client4 lost 1 out of 4
client4 duplicated 0
client4 out of order 0
mean delay 7.06
max delay 9.19
min delay 4.36
variance delay 6.0759
mean skew 1
max skew 1
min skew 1
variance skew 0

Sensor temp_1
Server got 30 out of 30
client1 lost 0 out of 30
client1 duplicated 0
client1 out of order 0
mean delay 0.000666666
max delay 0.00999999
min delay 0
variance delay 6.436769e-06
mean skew 0
max skew 0
min skew 0
variance skew 0
client2 lost 0 out of 30
client2 duplicated 0
client2 out of order 0
mean delay 0.000666666
max delay 0.00999999
min delay 0
variance delay 6.436769e-06
mean skew 0

```

max skew 0
min skew 0
variance skew 0
client3 lost 1 out of 30
client3 duplicated 0
client3 out of order 0
mean delay 1.002069
max delay 1.01
min delay 1
variance delay 1.699504e-05
mean skew 1
max skew 1
min skew 1
variance skew 0
client4 lost 1 out of 30
client4 duplicated 0
client4 out of order 0
mean delay 1.002069
max delay 1.01
min delay 1
variance delay 1.699504e-05
mean skew 1
max skew 1
min skew 1
variance skew 0
-----
>
> proc.time()
      user  system elapsed
    0.176   0.032   0.199

```